Multistrip Stitching Imaging Mission Planning Method for SAR Satellite Regional Mapping Considering Onboard Energy Consumption

Xin Shen, Zezhong Lu, Litao Li, Yaxin Chen, Xufei Li, Jiaying Wang, and Wei Yao

***Abstract*— To meet the demands of regional mapping with synthetic aperture radar (SAR) satellites, this study proposes a multistrip stitching imaging mission planning (MSIMP) method that considers energy consumption. Accounting for both regional coverage rate and satellite energy consumption, an MSIMP model was constructed. The model maximizes the region coverage benefit (RCB), minimizes the total imaging time (TIT) as the optimization objectives, and sets the indices of orbit selection, side-swing angle coefficients, and determination coefficients of the start time and end times of each orbit as the decision variables to realize the overall optimal scheduling of available orbital resources (ORs), side-swing angles, and imaging time of satellites. To address the problem of the large-scale and diverse types of decision variables in the MSIMP model, an improved particle swarm optimization (PSO) algorithm with a hybrid local search and differential evolution (LSDE) strategy (LSDE-PSO) that implements differential operations and local searches on two types of particles at various stages of the evolution process to enhance the model's solution efficiency was proposed. The proposed method was validated using three simulation scenarios with different regional sizes. The experimental results showed that the proposed method demonstrated consistent adaptability to different-scale tasks, achieving the synchronized optimization of coverage revenue and on-orbit energy consumption. Compared with existing algorithms, the proposed LSDE-PSO algorithm can obtain the optimal imaging scheme with a higher RCB with lower TIT at almost the same computational cost, which provides significant technical support for mission planning for regional SAR satellite imaging in practical applications.**

***Index Terms*— Imaging mission planning (IMP), regional mapping, synthetic aperture radar (SAR) satellites, stitching imaging, synchronized optimization.**

I. Introduction

REMOTE sensing satellite regional mapping has played an important role in an increasing number of fields, such as land census and monitoring [1], emergency response and rescue [2], and urbanization analysis and urban expansion [3]. Owing to the advantages of all-time and all-weather imaging, synthetic aperture radar (SAR) satellites have become an important means of acquiring high-resolution images in large region mapping [4], [5].

During regional high-resolution image acquisition, the imaging width of mainstream SAR satellites is relatively narrow. For example, TerraSAR has a 30-km imaging width in the stripmap mode (3-m resolution) and 10-km imaging width in the spotlight model (1-m resolution) [6], RadarSat-2 has an 18-km imaging width in the spotlight model [7], and Gaofen-3 has a 30-km imaging width in the hyfine strip mode (3-m resolution) [8]. Therefore, SAR satellites achieve the complete coverage of large regions through the stitching of imaging strips by multiorbit transit, termed multistrip stitching imaging. Notably, compared with the passive remote sensing mode of optical satellites, the peak power consumption of active imaging of SAR satellites is higher [9], which puts forward high requirements for on-orbit energy guarantee and imaging timing selection of the entire satellite (see Fig. 1). Therefore, the process of region mapping for SAR satellites necessitates the development of a scientific multistrip stitching imaging scheme to determine the appropriate satellite, amount of time (transit circle number, and imaging start and end times), and side-swing angle to image the region to achieve complete coverage of the task with minimum onboard energy consumption (shortest overall imaging time), ensuring the efficiency of the satellite's on-orbit work.

Given this context, this study aimed to propose a multistrip stitching imaging mission planning (MSIMP) method for the regional mapping of SAR satellites considering onboard energy consumption. The MSIMP model and the corresponding solution algorithm [an improved particle swarm optimization (PSO) algorithm with a hybrid local search and differential evolution (LSDE) strategy (LSDE-PSO)] achieve synchronous optimization of the selection of orbital resources (ORs) of multiple satellites, determination of the side-swing angle, and imaging duration. Therefore, the proposed method can effectively reduce the energy consumption of satellites with optimal coverage.

The main contributions of this study are given as follows.

1) The proposed mission planning method can reduce the energy consumption of satellites in the execution of imaging tasks under the premise of ensuring mission

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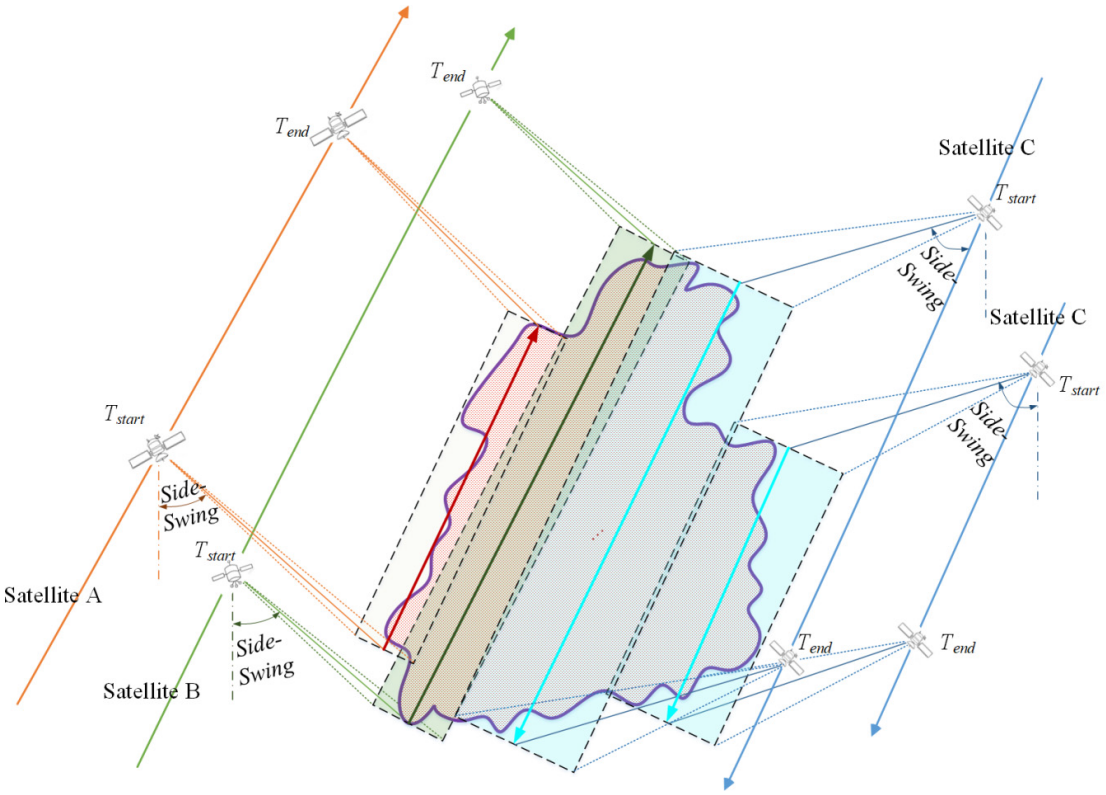


Fig. 1. Multistrip stitching imaging mission of SAR satellites.

coverage benefit, effectively improving the overall utilization efficiency of on-orbit satellite resources.

2) The proposed MSIMP model, which maximizes the region coverage benefit (RCB) as the primary optimization objective and minimizes the total imaging time (TIT) as the secondary optimization objective, under four types of decision variables, can realize the synchronous optimization of the imaging orbits, side-swing angle, and imaging start and end times. The model can achieve the integrated scheduling of multisatellite allocation, multiorbit resource, and imaging action, overcoming the limitations of the traditional method (first decomposition of regional target and then mission planning), which is difficult to achieve global optimization and is suitable for scheduling modeling of regional imaging of single-satellite/multisatellite.
3) The proposed LSDE-PSO algorithm can address the challenges posed by large-scale and diverse types of decision variables and large overhead calculations in the imaging mission planning (IMP) model of multiorbit splicing for a region. It can use two types of particles to implement different differential evolution and local search strategies, which improves the searchability of the algorithm and obtains an efficient imaging mission scheme with almost the same computing consumption.

The following sections of this study are organized as follows. In Section II, the details of constructing the MSIMP model are introduced. In Section III, the proposed LSDE-PSO algorithm for solving the MSIMP model is discussed. In Section IV, the simulation results of the proposed mission planning method are presented, and their performance and adaptability are analyzed. In Section V, the conclusion is presented.

II. Related Work

Existing studies have shown how to construct and solve the IMP model in a limited time and obtain several feasible solutions that meet the constraints, which has become a general approach for solving the IMP problem of satellites [10], [11], [12], [13], [14]. The existing IMP modeling of regional targets mainly adopts the ideas of "first decomposition of regional target and then mission planning" and "integrated modeling of decomposition of regional target and mission planning," as described in the following.

1) *First decomposition of regional target and then mission planning:* The decomposition of the regional target divides the region into a set of several imaging strips that can be completely covered by satellites. Mission planning allocates satellite resources based on the imaging strips. For example, a partition method of the predefined reference system was used to decompose a regional target into several independent scenes according to the size of the standard image frame using horizontal strips perpendicular to the track direction. Subsequently, a planning model was constructed to achieve optimal coverage with the least number of scenes [15]. The fixed-width strip-division method, which was adopted to plan the Pléiades satellite, decomposes an entire region into parallel strips of fixed width by controlling

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the entire decomposition process using two parameters (direction and offset). On this basis, a constraint satisfaction model was constructed with maximizing coverage benefit as the objective function and selection of imaging strips, determination of imaging action sequence, and imaging time as the decision variables [16]. The regional target was divided into a set of imaging strips with overlapping edges using a partitioning method for remote sensing parameters. Then, a mathematical model for target optimization was built by maximizing the coverage benefit as the objective function, selecting the imaging strips, and determining the imaging start time as decision variables [17]. A set of equally spaced imaging angles of satellites was constructed according to the side-swing angle, and a set of candidate imaging strips was generated by the partition method of dynamic decomposition of the regional target. Subsequently, a mission planning model that maximizes the mission coverage as the optimization objective and the selection of imaging strips as the decision variable was generated [18]. After obtaining a set of imaging strips using the partition method of dynamic decomposition of the regional target, construct a mission planning model of the regional target to obtain the optimal coverage benefit in the fastest time [19].

2) *Integrated modeling of decomposition of regional target and mission planning:* The decomposition of regional target and the allocation of satellite resources can be performed simultaneously during mission planning. For example, for the imaging requirement of the regional target of the lunar surface, a method based on the logic operation of a vector polygon was proposed to calculate the comprehensive coverage benefit of multiple nonparallel imaging strips. A side-swing angle optimization model for splicing imaging was constructed with the optimal coverage as the objective function and imaging side-swing angle as the decision variable [20]. To solve the problem of regional mapping, a method based on the Vatti algorithm was proposed to calculate the benefit of multiple nonparallel imaging strips. A multiobjective IMP model was constructed, which takes maximizing the coverage benefit and minimizing the number of ORs of the satellite as the optimization objectives and selection of ORs and side-swing angle as the decision variables [21], [22].

Existing optimization algorithms for solving planning models can be summarized as exact [23], [24], heuristic [11], [25], and machine learning [22], [26] algorithms. However, the heuristic algorithms remain the mainstream algorithms for solving planning models, owing to their superior generality and solving efficiency [12]. Existing heuristic algorithms have been mainly improved in three aspects: evolutionary operation, parameter adaptation, and combination with exact algorithms.

1) *Heuristic Algorithms of Improved Evolutionary Operation:* A decimal coding operator of the discrete PSO algorithm was designed, and a discrete PSO algorithm with a mutation operator was proposed to solve the planning model to avoid falling into the local optima of the algorithm [27]. An improved simulated annealing heuristic algorithm based on a fast insertion strategy was proposed to obtain the IMP scheme under cloud uncertainty [28]. A resampling PSO algorithm based on the initialization of a greedy strategy was proposed to generate an effective and efficient solution [29].

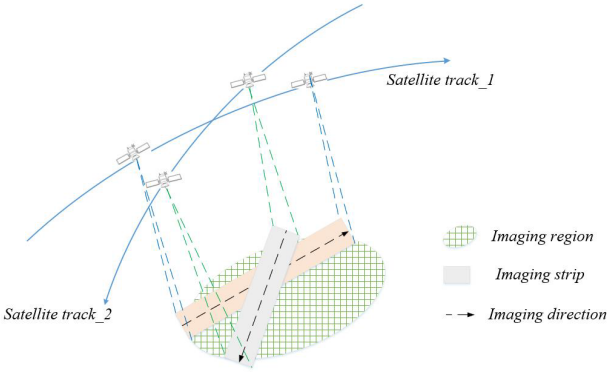


Fig. 2. Strip crossing of an SAR satellite.

2) *Heuristic Algorithms of Parameter Adaptation:* An adaptive simulated annealing algorithm with adaptive temperature control, short-term revisit avoidance mechanism based on the tabu list, and intelligent combination of neighborhood structure was proposed by Gu et al. [30]. By introducing adaptive probability into crossover and mutation operators and adding the optimization idea of the tabu search algorithm, an improved hybrid genetic algorithm was proposed to solve the planning problem, improving the probability of searching for the global optimal solution and accelerating the convergence rate of the algorithm [31]. An improved adaptive large-domain search algorithm with two stages of task assignment and task ranking was proposed to solve the planning model [25].

3) *Combination of Heuristic and Exact Algorithms:* An effective hybrid optimization method based on the combination of PSO and a genetic algorithm was proposed. This method used a genetic crossover operator to form a new generation of particles and realized a balance of local solutions and global solutions [32]. A hybrid algorithm based on PSO and gravitational search algorithms was proposed to improve the local search ability and reduce the convergence speed of the algorithm [33]. A two-stage hybrid planning algorithm was proposed; first, an improved firework algorithm based on the disturbance and update strategy of the population was used to determine the indivisible task sequence, and then, a heuristic algorithm was used to plan the divisible task sequence [11].

According to the comprehensive analysis of the above related literature, the current research on IMP methods has mainly focused on reducing ORs of satellites under the premise of optimal coverage; meanwhile, the synchronous implementation of coverage benefits and the energy consumption is still in the initial stage. The modeling idea of first

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decomposing the regional target and then planning the mission raises the problem that the result of mission planning is closely related to the results of task decomposition, which severely restricts the acquisition of an "optimal" imaging scheme. In addition, for the multisatellite imaging mission, the modeling idea faces the problem of satellite resource waste from the nonparallel orbit among satellites; in particular, the same SAR satellite has a strip crossing during ascending and descending orbits (see Fig. 2). The strip crossing of multiple SAR satellite imaging is more evident than that of multiple optical satellite imaging. The integrated modeling method primarily maximizes the coverage benefit and minimizes the number of ORs as optimization objectives [21]. Because the imaging duration is not considered an optimization objective in the planning model, the energy consumption of the satellites cannot be reduced. As the integrated modeling must simultaneously optimize the decomposition of the regional target and allocation of satellite resources, the complexity and computational consumption of model solving further increase. Moreover, the decision variables of the planning model have large dimensions and diverse types, which significantly reduces the ability of the heuristic algorithm to find the optimal solution. Therefore, this necessitates the development of an IMP method (planning model and solution algorithm) that considers the regional coverage and energy consumption of satellites.

III. Method

This section discusses the MSIMP model based on the introduction of decision variables. Our goal is to achieve synchronous optimization of the ORs of satellites, side-swing angle, and imaging duration while ensuring optimal coverage benefit with minimal energy consumption.

A. Model Hypothesis

In this study, the MSIMP problem, considering the satellite's coverage benefit and energy consumption, was transformed into an optimization problem related to the selection of ORs, determination of the side-swing angle, and imaging start and end times. Furthermore, we made the following assumptions: 1) the imaging task of satellites is a regional target; 2) new insertion tasks are not considered during the imaging process; 3) the constraint conditions of the model mainly consider the constraint of the side-swing angle; and 4) a sufficient amount of energy and storage space is available on the satellites.

B. Determination of the Decision Variables

The MSIMP of SAR satellites typically focuses on optimizing the side-swing angle of the satellites to build a planning model. The relevant literature has proposed a planning model to achieve optimal coverage with a minimum number of ORs [21]. However, they did not optimize the mission imaging duration. Therefore, although an imaging scheme with fewer ORs can be obtained, synchronous optimization of the coverage benefit and energy consumption of the satellites cannot be achieved. To solve this problem, we introduced four kinds of decision variables.

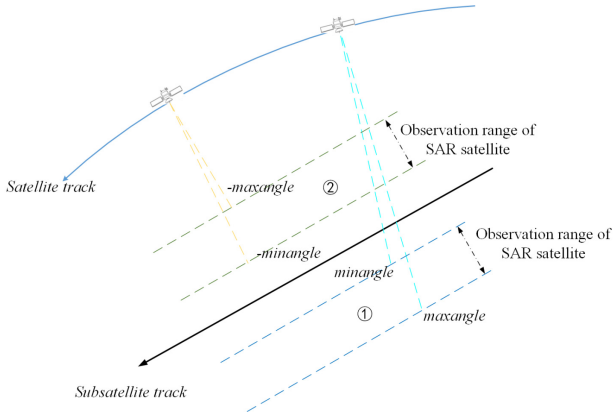


Fig. 3. Observation range of an SAR satellite.

1) Indices of selection of ORs (G_{ij}): This decision variable selects the ORs of the satellites participating in imaging from a preprocessed set of satellite orbits. G_{ij} is encoded as a binary number of 0 or 1, where 1 indicates that the orbit is selected to participate in the imaging of the satellite and 0 indicates that the orbit is not involved in satellite imaging

$$G_{ij} \in \{0, 1\} \quad \forall i \in \{1, 2, \ldots, n\}, \ \ j \in \{1, 2, \ldots, m_i\} \tag{1}$$

where i represents the ith satellite, n is the number of satellites, j indicates the jth orbit of the ith satellite, and m_i indicates the number of orbits obtained by the preprocessing of the ith satellite.

2) Side-Swing Angle Coefficient (R_{ij}): During the imaging process of the SAR satellites, only the area within the range of the left and right sides of the side-swing angle is captured (see Fig. 3). Therefore, if the side-swing angle is directly taken as the decision variable of the MSIMP model, defining the value range of the iterative operation will be difficult when using the optimization algorithm because of the existence of two side-swing angle ranges. The side-swing angle coefficient is, thus, introduced as a decision variable in the MSIMP model to restore the side-swing angle of an actual satellite

$$R_{ij} \in [\min_i, \max_i] \quad \forall i \in \{1, 2, \ldots, n\}, \ \ j \in \{1, 2, \ldots, m_i\} \tag{2}$$

where R_{ij} represents the side-swing angle coefficient of the jth orbit of the ith satellite, $[\min_i, \max_i]$ indicates the value range of the side-swing angle coefficient of the ith satellite, and $\max_i$ and $\min_i$ represent the maximum and minimum values of R_{ij}, respectively. The specific values are given as follows:

$$\begin{aligned} \max_i &= \text{maxangle}_i - \text{minangle}_i \\ \min_i &= \text{minangle}_i - \text{maxangle}_i \end{aligned} \tag{3}$$

where maxangle_i and minangle_i indicate the maximum and minimum side-swing angles of the ith satellite, respectively.

The MSIMP model can restore the actual side-swing angle during satellite imaging according to R_{ij}, and the calculation formula is shown as follows:

$$\begin{aligned} \text{if } R_{ij} \geq 0 \text{ then } \text{Swing}_{ij} &= R_{ij} + \text{minangle}_i \\ \text{else } \text{Swing}_{ij} &= R_{ij} - \text{minangle}_i \end{aligned} \tag{4}$$

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where Swing_{ij} represents the actual side-swing angle of satellite imaging obtained by R_{ij}.

3) Determination Coefficients of Imaging Start and end Times (t_{ij}^s and t_{ij}^e): These are used to restore the actual start and end times of target imaging under the current satellite orbit and satellite side-swing angle determined by the decision variables G_{ij} and R_{ij}

$$\begin{aligned} t_{ij}^s &\in \left[-(TW_{ij}^e - TW_{ij}^s)/2, TW_{ij}^e - TW_{ij}^s\right] \\ &\quad \forall i \in \{1,2,\ldots,n\}, \quad j \in \{1,2,\ldots,m_i\} \\ t_{ij}^e &\in \left[-\left(TW_e^{ij} - TW_s^{ij}\right), \left(TW_e^{ij} - TW_s^{ij}\right)/2\right] \\ &\quad \forall i \in \{1,2,\ldots,n\}, \quad j \in \{1,2,\ldots,m_i\} \end{aligned} \tag{5}$$

where t_{ij}^e and t_{ij}^s represent the determination coefficients of the end and start times of the jth orbit of the ith satellite, respectively, and TW_{ij}^e and TW_{ij}^s indicate the end and start times of the jth orbital window of the ith satellite obtained by preprocessing, respectively.

The start (T_{ij}^s) and end (T_{ij}^e) times of satellite imaging can be restored according to (t_{ij}^s and t_{ij}^e), and the calculation formulas are shown as follows:

$$\begin{aligned} T_{ij}^s &= t_{ij}^s + TW_{ij}^s \\ T_{ij}^e &= t_{ij}^e + TW_{ij}^e. \end{aligned} \tag{6}$$

Therefore, we can use t_{ij}^s and t_{ij}^e to jointly confirm the imaging duration of the satellite in the current orbit to reduce energy consumption and improve imaging efficiency.

C. Objective Functions and Constraint Conditions

The MSIMP model contains two optimization objectives (maximizing the RCB and minimizing the TIT) and one constraint (side-swing angle), which aims to ensure that satellites can achieve optimal coverage of regional target with minimal energy consumption

$$\max \frac{S_{\text{cov}}\left(G_{ij}, R_{ij}, t_{ij}^s, t_{ij}^e\right)}{S_{\text{obj}}} \tag{7}$$

$$\min \sum_{i=1}^{n} \sum_{j=1}^{m_i} G_{ij} * \left(T_{ij}^e - T_{ij}^s\right) \tag{8}$$

$$\begin{aligned} \text{Swing}_{ij} &\in [\text{minangle}_i, \text{maxangle}_i] \\ &\quad \cup [-\text{maxangle}_i, -\text{minangle}_i]. \end{aligned} \tag{9}$$

The objective function (7) maximizes the RCB. $S_{\text{cov}}(G_{ij}, R_{ij}, t_{ij}^s, t_{ij}^e)$ indicates the effective coverage area of the imaging strip set determined by the decision variables for the regional target, and S_{obj} represents the total area of the regional target. The specific calculation process can refer to relevant literature [19]. The objective function (8) minimizes the TIT, and G_{ij} is the index of the selection of ORs. $G_{ij} = 1$ indicates that the jth orbit of the ith satellite is involved in the calculating the coverage benefit. $T_{ij}^e - T_{ij}^s$ represents the imaging duration of the jth orbit of the ith satellite. Equation (9) is the constraint of the side-swing angle, which indicates that the side-swing angle obtained by the reduction of the decision variables should be within the constraints of the maximum and minimum side-swing angles under the side-swing condition of the satellite itself. Note that the MSIMP model is a prioritization ranking model, in which objective function (7) is the primary optimization objective and objective function (8) is the secondary optimization objective, aiming to obtain the planning scheme with the shortest mission imaging duration under the premise of maximizing the RCB.

IV. Model Solving Algorithm

This section discusses the construction of the LSDE-PSO algorithm to solve the MSIMP model presented in Section II. Since the IMP problem is a nonpolynomial time complete problem, a variety of heuristic algorithms with global search capabilities, such as iwPSO [34], GA [35], and DE [36], have been used to solve the IMP problem to obtain a satisfactory planning scheme.

The standard PSO algorithm emphasizes that all particles in the population learn from the position of the global optimal particle but ignores the learning among particles, which easily falls into the local optimal solution and reduces the exploration ability of the algorithm. Therefore, the LSDE-PSO algorithm was proposed.

In contrast to the learning strategy of the standard PSO algorithm, we adopted a new learning strategy (see Fig. 4).

1) Two different particles in the population were randomly selected, and the win and lose particles were distinguished according to their fitness values.
2) The win particle was updated using the hybrid LSDE strategy to generate the winner particle, the specific updating process of which is shown in Fig. 5.
   Based on the particularity of the MSIMP model, the algorithm first determines whether the win particle should be updated to generate the winner particle according to the following equation:

$$\begin{aligned} &\text{if } S > a_1, \quad \text{then update} \\ &\text{else } S \le a_1, \quad x_{\text{winner},G}^d = x_{\text{win},G}^d \end{aligned} \tag{10}$$

   where a_1 is a random number within the range [0, 1], S represents the update probability, which is related to the times of iteration, $S = 1 - G/\text{max_gen}$, G indicates the Gth iteration, and max_gen is the maximum times of iteration. Equation (10) indicates that the win particle has a high probability of updating at the early stage of the iteration, which is conducive to improving the diversity and enhancing the exploration ability of the algorithm. The update probability of the win particle in the late iterations is small, which is conducive to the overall mining ability of the algorithm.
   Subsequently, according to (11), we judged whether a local search or differential evolution was used to update the win particle to produce the winner particle. When the RCB of the win particle is less than 100%, the winner particle is generated by the local search method. When the RCB of the win particle is equal to 100%, the winner

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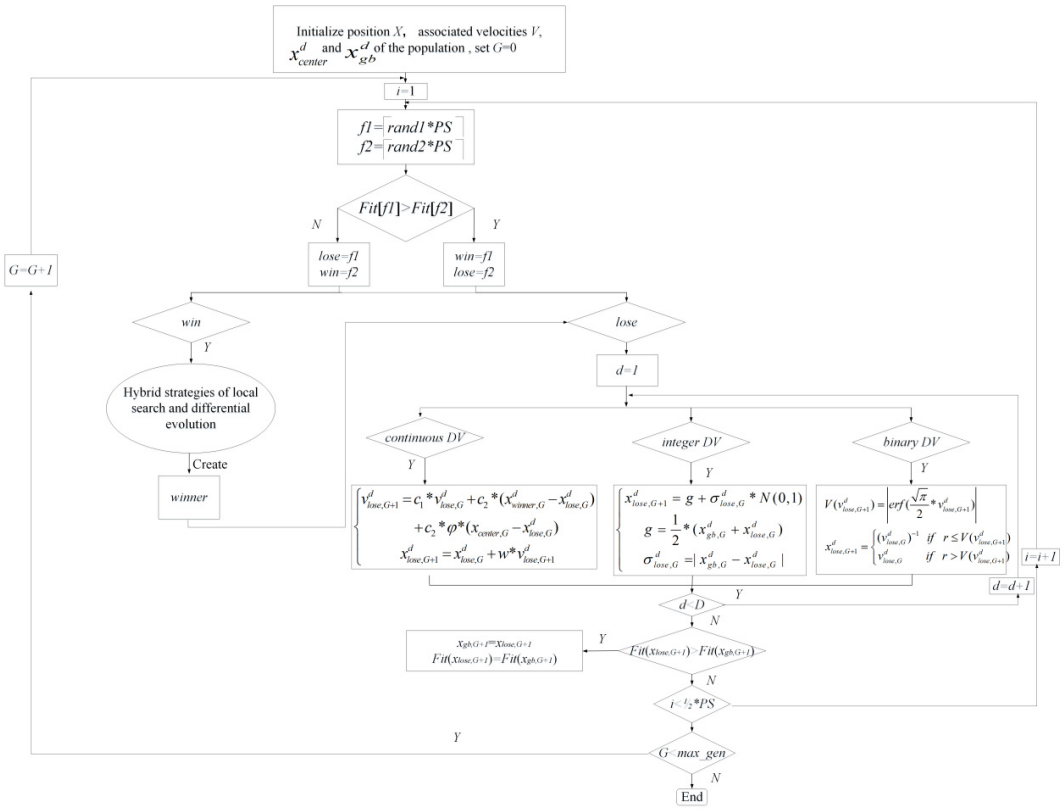


Fig. 4. Improved PSO algorithm with a hybrid LSDE strategy.

particle is generated by the differential evolution method

$$\begin{aligned}&\text{if } \mathrm{Fit_{win}}(\text{coverage}) < 1, \quad \text{perform local search}\\ &\text{else } \mathrm{Fit_{win}}(\text{coverage}) = 1,\\ &\qquad\qquad \text{perform differential evolution.} \end{aligned} \tag{11}$$

$\mathrm{Fit_{win}}$(coverage) represents the RCB of the current win particle.

We updated only the binary (G_{ij}) and discrete (t_{ij}^{s} and t_{ij}^{e}) decision variables while generating the winner particle using the differential evolution method.

The differential evolution method generates the winner particle as follows.

A differential strategy was used to mutate the win particle. Two different particles in the population were randomly selected, and the vector difference between the two particles was scaled to perform vector synthesis with the win particle, as shown in the following:

$$x_{\mathrm{winner},G}^{d} = x_{\mathrm{win},G}^{d} + F\left(x_{r1,G}^{d} - x_{r2,G}^{d}\right) \tag{12}$$

where $r1$ and $r2$ are two random integers with values in the range [1, PS], PS is the population size, and $d = 1, 2, \ldots, D$ represents the dimensions of the vector. F is the scaling factor, G indicates the Gth iteration, $x_{\mathrm{win},G}^{d}$ represents the position of the win particle in the Gth iteration, and $x_{\mathrm{winner},G}^{d}$ indicates the winner particle in the Gth iteration obtained by the win particle through the mutation operation.

A cross-operation was performed on the winner particle obtained by the mutation operation to determine the position of the winner particle after crossing, as shown in the following equation:

$$\begin{aligned}&\text{if rand()} <= \mathrm{CR}, \quad x_{\mathrm{winner},G}^{d} = x_{\mathrm{winner},G}^{d}\\ &\text{else rand()} > \mathrm{CR}, \quad x_{\mathrm{winner},G}^{d} = x_{\mathrm{win},G}^{d}\end{aligned} \tag{13}$$

where rand() represents a random number in the range [0, 1] and CR is a crossover operator with a value range of [0.95, 1], which is used to control the selection of the position of the winner particle after mutation or original win particle.

According to (14), the win and winner particles were compared based on their respective fitness values to determine the position of the final winner particle

$$\begin{aligned}&\text{if Fit(winner)} \geq \text{Fit(win)}, \quad x_{\mathrm{winner},G}^{d} = x_{\mathrm{winner},G}^{d}\\ &\text{else Fit(winner)} < \text{Fit(win)}, \quad x_{\mathrm{winner},G}^{d} = x_{\mathrm{win},G}^{d}.\end{aligned} \tag{14}$$

The local search method generates the winner particle as follows.

For G_{ij}, (15) was used to update, and for t_{ij}^{s} and t_{ij}^{e}, (16) was used to update. Meanwhile, two random numbers, a_2 and a_3, were introduced for comparison with the update probability S_1 to determine whether to update the decision variable. The value ranges of a_2 and a_3 were [0, 1], and S_1 was selected within the range of [0, 1]. In this study, we chose $S_1 = 0.5$, indicating that the decision variable had a 50% probability of updating

$$\text{if} \quad x_{\mathrm{win},G}^{d} = 0, \quad x_{\mathrm{winner},G}^{d} = 1 \tag{15}$$

$$x_{\mathrm{winner},G}^{d} = x_{\mathrm{win},G}^{d} + \mathrm{int}(\mathrm{random}[-1, 1]). \tag{16}$$

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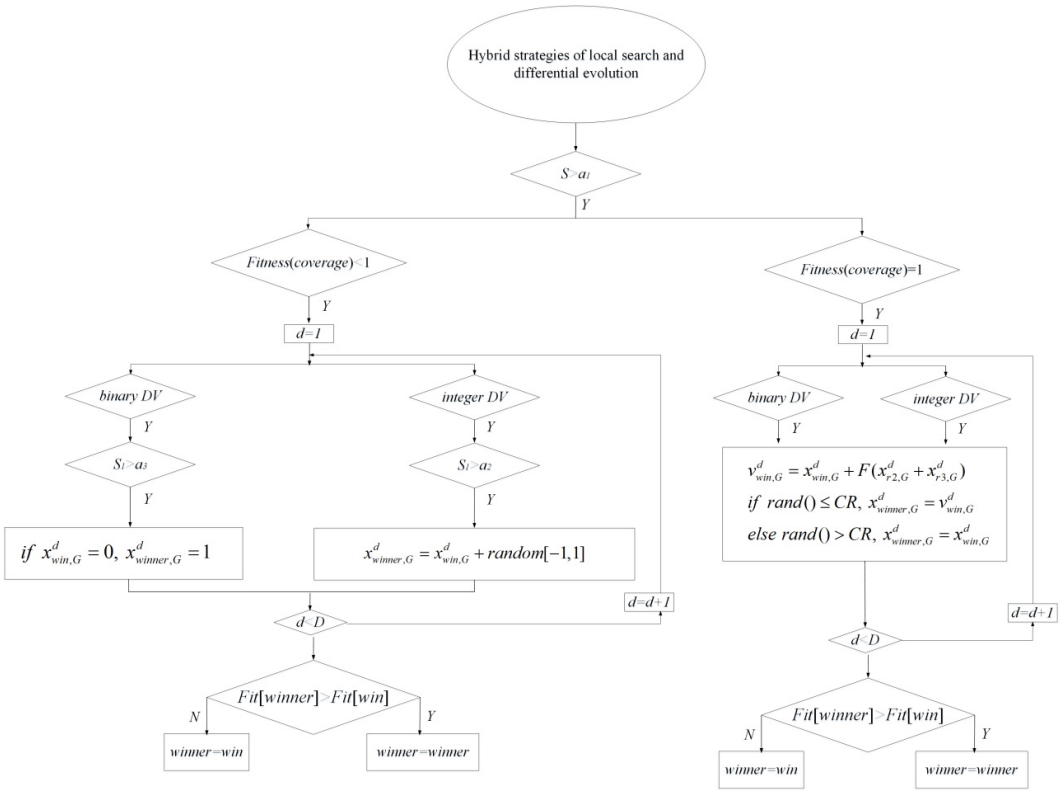


Fig. 5. Hybrid LSDE strategy.

3) The lose particle was updated considering the position of the updated winner particle and the average position of all particles as the evolutionary direction. On this basis, different updating paradigms were adopted for different types of decision variables.

For the continuous decision variable (R_{ij}), (17) was used to update [37]

$$\begin{cases} v^d_{\text{lose},G+1} = c_1 * v^d_{\text{lose},G} + c_2 * \left(x^d_{\text{winner},G} - x^d_{\text{lose},G}\right) \\ +c_3 * \varphi * \left(x^d_{\text{center},G} - x^d_{\text{lose},G}\right) \\ x^d_{\text{lose},G+1} = x^d_{\text{lose},G} + w * v^d_{\text{lose},G+1} \end{cases} \tag{17}$$

where $v^d_{\text{lose},G}$ represents the speed of lose particle in the Gth iteration, $x^d_{\text{winner},G}$ indicates the position of the winner particle generated by the hybrid LSDE strategy in the Gth iteration, $x^d_{\text{lose},G}$ is the position of lose particle in the Gth iteration, $x^d_{\text{center},G}$ represents the average position of all particles in the Gth iteration, c_1, c_2, and $c_3 \in [0, 1]$ are the acceleration factors, w is the inertia weight, and φ represents the optimal average weight.

For the discrete decision variable (G_{ij}), (18) was used to update [38]

$$\begin{cases} x^d_{\text{lose},G+1} = g + \sigma^d_{\text{lose},G} * N(0, 1) \\ g = \dfrac{1}{2} * \left(x^d_{gb,G} + x^d_{\text{lose},G}\right) \\ \sigma^d_{\text{lose},G} = \left|x^d_{gb,G} - x^d_{\text{lose},G}\right| \end{cases} \tag{18}$$

where $N(0, 1)$ is the Gaussian normal distribution and $x^d_{gb,G}$ represents the optimal position of all particles in the Gth iteration.

In the study of S- and V-type mapping functions, Mirjalili and Lewis [39] found that the S-type mapping function causes the optimization algorithm to fall into local minima and slow convergence speed. In contrast, the V-type mapping function can effectively circumvent this problem. Therefore, for binary decision variables (t^s_{ij} and t^e_{ij}), we used a V-type mapping function to update the position, where the updated paradigm of speed was the same as that of (17). The updating paradigm for the position is shown in the following equation:

$$\begin{aligned} V\left(v^d_{\text{loser},G+1}\right) &= \left|\text{erf}\left(\frac{\sqrt{\pi}}{2} v^d_{\text{loser},G+1}\right)\right| \\ x^d_{\text{loser},G+1} &= \begin{cases} \left(v^d_{\text{loser},G}\right)^{-1}, & \text{if } r \le V\left(v^d_{\text{loser},G+1}\right) \\ v^d_{\text{loser},G}, & \text{if } r > V\left(v^d_{\text{loser},G+1}\right) \end{cases} \end{aligned} \tag{19}$$

where $V(v^d_{\text{lose},G+1})$ denotes a V-type mapping function and $r \in [0, 1]$ is a random value.

The specific process of the LSDE-PSO algorithm is given as follows.

1) Initialize the position and speed of each particle in the population, calculate the global optimal position and average position of all particles, and set the population size and the maximum times of iterations.
2) Randomly select two particles from the population, and compare them based on their fitness values to determine the win and lose particles.
3) Use the hybrid LSDE strategy to update the win particle and generate the winner particle.

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TABLE I
ORBITAL PARAMETERS OF EACH SATELLITE

Sat ID	Sat_1	Sat_2
Semi-major Axis (*m*)	6877200.317	6905596.741
Eccentricity	0.002319	0.001441
Inclination (*deg*)	97.31	97.412
RAAN (*deg*)	259.08	236.586
Arg of Perigee (*deg*)	70.438	313.984
Mean Anomaly (*deg*)	159.385	159.159

TABLE II
SENSOR PARAMETER OF EACH SATELLITE

Sat ID	Sat_1	Sat_2
Angle of Incidence (*deg*)	22°~24°	15°~40°
Width of Strip (*km*)	20	25
Vision of Right and Left	Support	Support

4) Use the updated winner particle and updated paradigm of different types of decision variables to update the lose particle.
5) Compare the fitness value of the updated lose particle with the fitness value of the global optimal position under the condition of "un-meeting the iteration termination condition."

V. EXPERIMENT AND ANALYSIS

To verify the performance and applicability of the proposed MSIMP method for the regional mapping of SAR satellites considering onboard energy consumption, two satellite parameters were set to conduct simulation imaging experiments on three regions: the sources of the Lancang River, the Yellow River, and the Yangtze River in Tibet.

A. Experimental Setting

The orbit and sensor parameters for each satellite in the experimental verification process are listed in Tables I and II, respectively. The specific distributions of the three regions are shown in Fig. 6, and the area size of each source, the imaging period, and the number of preprocessed ORs used for planning are listed in Table III.

B. Experimental Results

To verify the effectiveness of the proposed method, a comparative experiment was designed using two levels of models and algorithms.

1) Model Comparison Experiment: In the comparison experiment, the MSIMP model was compared with the two other models. Model A is a planning model that achieves optimal coverage with the least number of ORs [21] and Model B is a greedy model that realizes optimal coverage with minimal energy consumption (optimizing only one orbit at a time, ensuring that the accumulation of all ORs obtained by preprocessing can obtain the maximum RCB and the minimum TIT).

After optimizing and solving the above three models, the planning results of the RCB, TIT, and optimized ORs (OORs) of each scenario were calculated, as shown in Table IV. The IMP scheme obtained by solving each model is shown in Fig. 7.

A comprehensive analysis of the planning results of the three models in the above three scenarios (see Table IV and Fig. 7) revealed that all three models can obtain 100% RCB when the scale of the imaging task was small (Scenarios A and B). Under the premise of obtaining 100% RCB, the TIT of the MSIMP model was shorter than that of Models A and B. Compared with Models A and B, the TIT of the MSIMP model decreased by 63.82% and 20.3% in Scenario A and 55.91% and 29.63% in Scenario B, respectively. Meanwhile, the planning scheme obtained using the MSIMP model also had the least number of OORs. The numbers of OORs saved by each model in Scenario A were 50%, 21.43%, and 50%, and those in Scenario B were 45.83%, 20.83%, and 54.17%, respectively. When the scale of the imaging task was large (Scenario C), only Model A and the MSIMP model achieved 100% RCB. However, the planning scheme of the MSIMP model requires a shorter TIT and fewer OORs. Compared with Model A, the TIT of the proposed MSIMP model decreased by 69.05%, and the OORs saved by the two models were 16.13% and 29.03% respectively.

By analyzing the model comparison results, the following conclusion can be drawn.

The MSIMP model can obtain better RCB and TIT and less OORs and simultaneously realize the decision of coverage benefit and energy consumption of the satellite (selection of ORs, determination of side-swing angle, and imaging duration).

The MSIMP model is suitable for imaging tasks at different scales.

2) Algorithm Comparison Experiment: To verify the effectiveness of the proposed method, the iwPSO, GA, and DE were compared with the LSDE-PSO algorithm. When solving the algorithms, different population sizes and iteration times were adopted according to the area size of different scenarios (the dimension of the decision variable). The parameter configuration of the algorithms is shown in Table V. The above algorithms were used to optimize and solve each imaging region ten times, and the average optimization results of the RCB and TIT of each scenario are shown in Fig. 8. The average times consumed by each algorithm are listed in Table VI.

As shown in Fig. 8, at the level of the optimization effect, the order of the RCB obtained by the optimization solutions of various algorithms in the three scenarios was LSDE-PSO > iwPSO > DE > GA. Compared with the other three algorithms, only the LSDE-PSO algorithm achieved 100% RCB in the three scenarios, and the TIT was shorter than that of other algorithms. At the optimization efficiency level, the time consumption of the LSDE-PSO algorithm was almost similar to that of the other algorithms (see Table VI).

The LSDE-PSO algorithm can realize observation of regional scenarios with a higher RCB and shorter TIT, thereby demonstrating better performance than the other algorithms.

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TABLE III
SIZE OF EACH SOURCE

Imaging source	Area (km^2)	Imaging time period	Number of preprocessed ORs
Lancang River Source (Scenario A)	14871.6209	2022-06-20 08:00:00 to 2022-07-05 08:00:00	Sat_1: 10 orbits, Sat_2: 18 orbits
Yellow River Source (Scenario B)	20166.8517	2022-06-20 08:00:00 to 2022-07-05 08:00:00	Sat_1: 9 orbits, Sat_2: 15 orbits
Yangtze River Source (Scenario C)	91526.8317	2022-06-20 08:00:00 to 2022-07-15 08:00:00	Sat_1: 26 orbits, Sat_2: 36 orbits



(a) (b) (c)

Fig. 6. Imaging source of satellite. (a) Lancang River. (b) Yellow River. (c) Yangtze River.

TABLE IV
PLANNING RESULT OF RCB, TIT, AND OR FOR EACH MODEL

Scenarios	Model A			Model B			Propose model		
	RCB (%)	TIT (s)	OORs	RCB (%)	TIT (s)	OORs	RCB (%)	TIT (s)	OORs
Scenario A	100	434	14	100	197	22	100	157	14
Scenario B	100	431	13	100	270	19	100	190	11
Scenario C	100	2814	52	99.65	1200	59	100	871	44

It also has better adaptability in solving scheduling problems considering both the regional coverage and energy consumption of satellites.

C. Experimental Analysis

1) The MSIMP model can synchronously optimize the ORs, side-swing angle, and imaging time. In the optimization process of Model A, although the obtained IMP scheme achieved higher coverage with the least ORs, the required number of ORs also increases as the scale of the task continues to increase. Meanwhile, as the TIT was not optimized, the imaging duration of the satellite was prolonged, which could easily lead to significant resource wastage as areas outside the regional target were also imaged. Model B was greedy optimizing only one OR each time in the optimization process to ensure that the orbit can obtain the optimal coverage benefit and the shortest imaging duration. Therefore, the IMP scheme obtained using Model B failed to consider the global situation and could not obtain the optimal scheme, resulting in a large number of ORs required for imaging. By introducing four types of variables as the decision variables (G_{ij}, R_{ij}, T_{ij}^{s}, and T_{ij}^{e}), the MSIMP model realized the integrated optimization of ORs, side-swing angle, and imaging duration through synchronous optimization of the decision variables and achieved complete coverage imaging of the task with fewer ORs and shorter TIT. The MSIMP model can improve the imaging efficiency and reduce the energy consumption of the SAR satellites.
2) The LSDE-PSO algorithm was superior to the other algorithms. The fundamental reason for its effective application to the MSIMP problem considering the coverage benefit and energy consumption is related to the learning strategy. The learning strategy adopted by the LSDE-PSO algorithm is no longer limited to learning from the historical and global optimal positions of the particles. The LSDE-PSO algorithm first uses the fitness values of the two particles to distinguish between the win and lose particles and then updates

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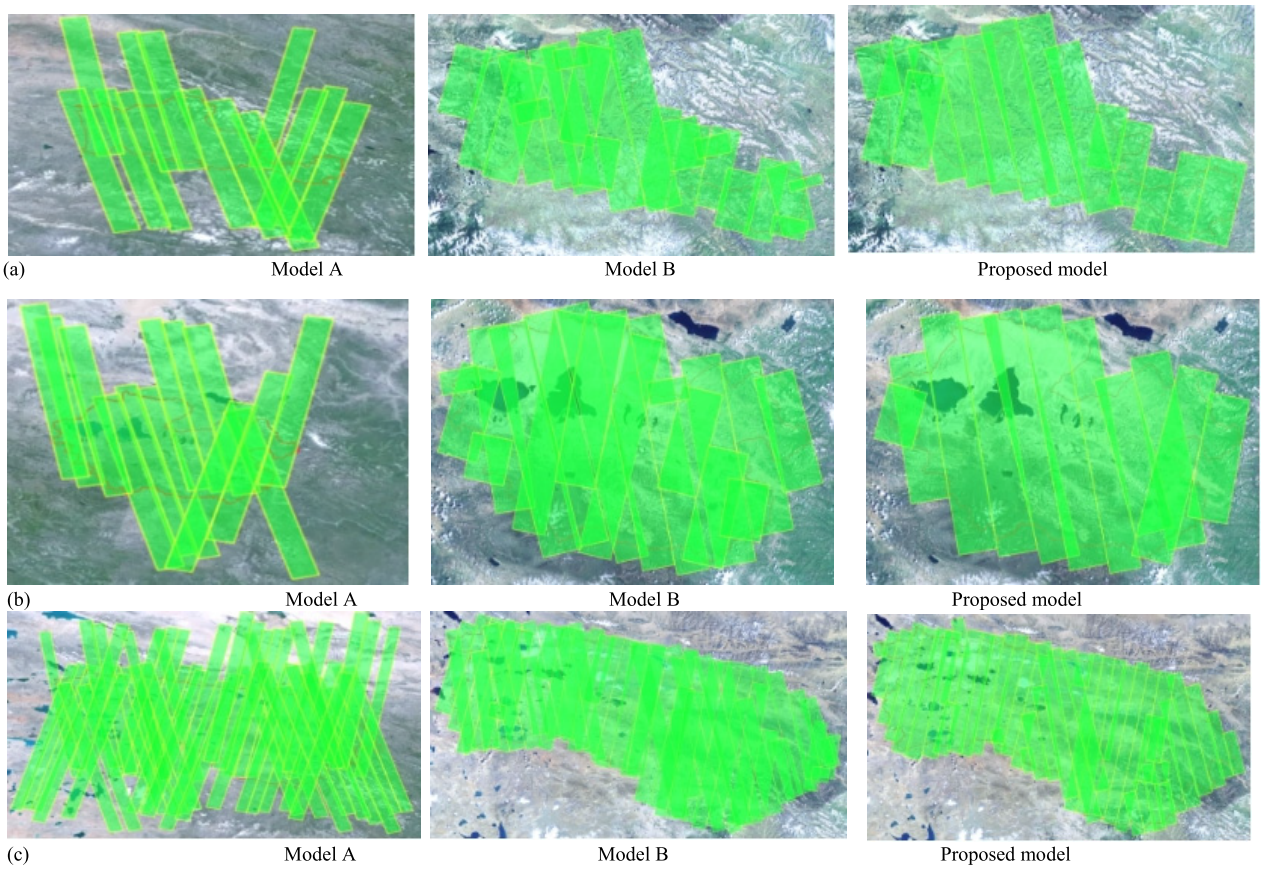


Fig. 7. IMP scheme of each model. (a) Scenario A. (b) Scenario B. (c) Scenario C.

TABLE V
PARAMETER CONFIGURATION OF EACH ALGORITHM IN EACH SCENARIO

Algorithms	Scenario	Parameter configuration	
iwPSO	A	$P = 200$, $IT = 2000$	$c1 = c2 = 2.0$, $w = 0.5$
	B	$P = 200$, $IT = 2000$	
	C	$P = 300$, $IT = 3000$	
GA	A	$P = 200$, $IT = 2000$	$F = 0.8$, $CR = 0.7$
	B	$P = 200$, $IT = 2000$	
	C	$P = 300$, $IT = 3000$	
DE	A	$P = 200$, $IT = 2000$	$F = [0.4,0.99]$, $CR = [0.0001,0.1]$
	B	$P = 200$, $IT = 2000$	
	C	$P = 300$, $IT = 3000$	
LSDE-PSO	A	$P = 200$, $IT = 2000$	$w = c1 = c2 = c3 = [0,1]$, $\varphi = 0.15$ $F = [0,1]$, $CR = [0.9,1]$
	B	$P = 200$, $IT = 2000$	
	C	$P = 300$, $IT = 3000$	

TABLE VI
AVERAGE TIME CONSUMPTION OF DIFFERENT ALGORITHMS IN EACH SCENARIO

Algorithm	Average time consumption (s)		
	Scenario A	Scenario B	Scenario C
iwPSO	42942.516	39665.808	118277.129
GA	43008.459	39760.49	118344.929
DE	43145.216	39905.752	118439.126
LSDE-PSO	42867.253	39634.395	118140.121

the win particle using a hybrid LSDE strategy to generate the winner particle. Finally, the position of the updated winner particle and the average position of the population are used to guide the update of the lose particle, which enhances the diversity of the population. Meanwhile, considering that the MSIMP model has various types of continuous-discrete decision variables, an adaptive updating strategy of decision variable types was introduced to improve the exploration ability of the algorithm. Therefore, compared with the other algorithms, the LSDE-PSO algorithm exhibited better adaptability. Moreover, the computational cost of the LSDE-PSO algorithm in solving the MSIMP model was primarily due to the time required to calculate the objective function. For iwPSO, DE, and GA, each particle of the population must be updated, and the calculated cost was $C = P * \mathrm{IT}$, where P represents the population size and IT is the number of iterations. In addition to updating half of the lose particles in the population, the LSDE-PSO algorithm

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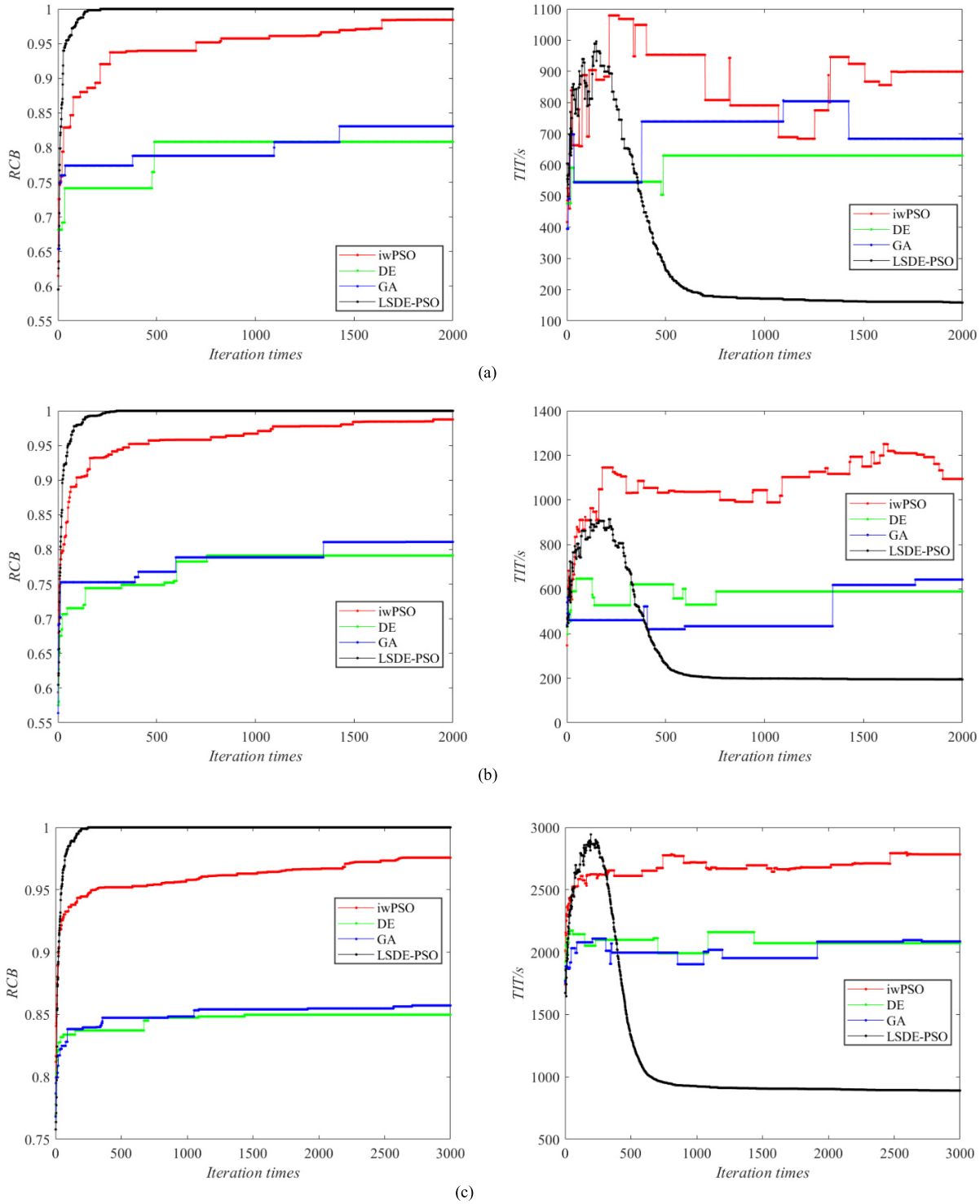


Fig. 8. Comparison results of RCB and TIT of each scenario under different algorithms. (a) Scenario A. (b) Scenario B. (c) Scenario C.

also needed to update half of the win particles to generate winner particles. Therefore, the calculation cost was $C = P * \mathrm{IT}$.

VI. Conclusion

To realize the imaging scheduling of multiple SAR satellites, we proposed an MSIMP method for the regional mapping of SAR satellites considering onboard energy consumption. First, we constructed an MSIMP model, which takes G_{ij}, R_{ij}, T_{ij}^{s}, and T_{ij}^{e} as the decision variables and maximizing the RCB and minimizing the TIT as the objective functions. Furthermore, to efficiently solve the MSIMP model, we proposed an improved PSO algorithm with a hybrid LSDE strategy. Finally, we designed three sets of imaging tasks for regional targets to verify the effectiveness of the proposed method. The experimental results showed that the proposed method

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can obtain an IMP scheme for SAR satellites with a higher RCB, shorter TIT, and fewer ORs.

In summary, the main findings of this study are given as follows.

1) The proposed MSIMP method for the regional mapping of SAR satellites considering onboard energy consumption can achieve higher coverage benefits with lower energy consumption of satellites (the number of OORs and orbit imaging duration) and effectively improve the utilization level of satellite resources.
2) The MSIMP model can synchronously realize the optimization of the ORs, side-swing angle, and imaging duration of satellites by constructing a quantitative relationship between decision variables and objective functions and ensuring the integrated scheduling of coverage benefit and energy consumption.
3) The proposed LSDE-PSO algorithm can obtain an imaging scheme based on a solution with a higher coverage benefit, lower TIT, and fewer orbit resources.

Future work will consider how to reasonably abstract the planning model to reduce the dimensions of the decision variables and the difficulty of solving. Furthermore, we will consider combining reinforcement learning and traditional algorithms to achieve a higher efficiency and higher quality solution for the planning model.

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